Bias Audit Summary (demonstration, not attested)

POTENTIAL ADVERSE IMPACT FOUND (impact ratio < 0.80)

System audited	HMDA mortgage decisions used as demo data for the NYC LL144 impact-ratio calculation (not an audit of any hiring tool)
Vendor / developer	Not specified
Tool distribution date	Not specified
Independent auditor	None (demonstration output, not an independent bias audit)
Audit date	2026-11-13
Data source	data/hmda_demo_sample.csv
Records analysed	8,767
Mode	selection

Standard and methodology

NYC Local Law 144 / EEOC four-fifths impact-ratio test. For each demographic category the selection rate (or, for scoring tools, the rate of scoring at or above the median) is computed, then the impact ratio = the category's rate divided by the highest category's rate. A ratio below 0.80 (four-fifths) is the long-standing EEOC threshold for potential adverse impact. Authority: 6 RCNY Sec. 5-300 et seq.; EEOC Uniform Guidelines 29 CFR 1607.4(D).

derived_race (min impact ratio: 0.6806) [POTENTIAL ADVERSE IMPACT]

Category	N	Rate	Impact ratio	< 0.80?
Native Hawaiian or Other Pacific Islander	11	0.545	0.610	excl. <2%
Black or African American	3265	0.609	0.681	YES
American Indian or Alaska Native	40	0.625	0.699	excl. <2%
2 or more minority races	52	0.750	0.838	excl. <2%
Asian	536	0.785	0.878	no
White	4455	0.844	0.944	no
Joint	408	0.895	1.000	no

derived_sex (min impact ratio: 0.7997) [POTENTIAL ADVERSE IMPACT]

Category	N	Rate	Impact ratio	< 0.80?
Male	3029	0.693	0.800	YES
Female	3258	0.723	0.835	no
Joint	2480	0.867	1.000	no

derived_race x derived_sex (min impact ratio: 0.6253) [POTENTIAL ADVERSE IMPACT]

Category	N	Rate	Impact ratio	< 0.80?
Native Hawaiian or Other Pacific Islander / Joint	1	0.000	0.000	excl. <2%
Native Hawaiian or Other Pacific Islander / Male	5	0.400	0.447	excl. <2%
American Indian or Alaska Native / Joint	2	0.500	0.559	excl. <2%
Black or African American / Male	1181	0.560	0.625	YES
2 or more minority races / Joint	5	0.600	0.670	excl. <2%
American Indian or Alaska Native / Female	25	0.600	0.670	excl. <2%
Black or African American / Female	1707	0.619	0.691	YES
Asian / Male	226	0.677	0.756	YES
2 or more minority races / Male	19	0.684	0.764	excl. <2%
American Indian or Alaska Native / Male	13	0.692	0.773	excl. <2%

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Black or African American / Joint	377	0.719	0.803	no
White / Male	1540	0.792	0.885	no
Native Hawaiian or Other Pacific Islander / Female	5	0.800	0.894	excl. <2%
2 or more minority races / Female	28	0.821	0.918	excl. <2%
White / Female	1276	0.842	0.940	no
Asian / Female	204	0.848	0.948	no
Joint / Joint	350	0.891	0.996	no
White / Joint	1639	0.895	1.000	no
Asian / Joint	106	0.896	1.001	excl. <2%
Joint / Male	45	0.911	1.018	excl. <2%
Joint / Female	13	0.923	1.031	excl. <2%

An impact ratio below 0.80 flags potential adverse impact and warrants further review; it is descriptive and is not by itself proof of unlawful discrimination. Computed only on the data provided; the audit itself simulates no records.

Evidence & attestation: NOT ATTESTED (demonstration only)

This output was generated without attestation. No ledger record was written, so there is no record hash, signature or public key to verify for it. The declared profile and the legal basis it was checked against are listed for reference only.

Audited as of 2026-11-13

Declared profile employment use in EU, US-NYC, US-CO, US-federal

Legal basis (point-in-time, source-hashed)

Obligation	Effective	Src	Source SHA-256 (12)
us-federal-nist-rmf	2023-01-26	CHANGED	c1de486bf233
us-nyc-ll144	2023-07-05	MANUAL	1f05401d2750